

SURENDER KANNAH | skannah@gmail.com | (832) 576 2297 | [LinkedIn](#) | [Github](#)

SUMMARY

Specialized in dense 3D perception, multi-camera vision systems, and robotics navigation using automotive-grade cameras. Built visual pipelines, stereo/depth systems, sensor fusion modules, and DRL-based autonomy controllers deployed on embedded hardware. NSF-funded engineer with experience converting video into structured world models.

TECHNICAL SKILLS

Computer Vision & 3D Perception – Stereo Vision, Depth Estimation, 3D Reconstruction, Point Clouds, Camera Calibration, YOLOv11, CNNs, Feature Detection, Image Segmentation, Optical Flow, PyTorch, OpenCV, CUDA, TensorRT

Autonomy & Robotics – Sensor Fusion, DRL Control Systems, High-fidelity simulation, EO/IR sensor processing, Embedded ML, Edge Deployment, Real-time systems, C++, Linux, ROS, Bayesian filtering, JPDA / MHT tracking, State estimation, Probabilistic data association, Stochastic processes, ODE/PDE modeling, nonlinear optimization

ML & Infrastructure – PINNs, tree-based models, ensemble methods, model optimization (8-bit quantization, pruning), evaluation, cross-validation, regularization, Python, C++, SQL, Docker, Git, Jupyter, VSCode, workflow automation

EDUCATION

University of Colorado at Boulder Master in Data Science May 2026
Relevant coursework: Machine Learning, Computer Vision, Data Mining, Algorithms

University of Texas at Austin Bachelor of Business Administration, Finance May 2019

RESEARCH EXPERIENCE

Autonomous Navigation System for Mobile Robots – Robotics Engineer | *Houston, Texas* 2026 - Present

- Built in Gazebo using TurtleBot3, Nav2, AMCL, and TF, achieving a 98% success rate across dynamic obstacle courses
- Integrated odometry plugins and configured global/local costmaps to reduce localization drift to 3 cm
- Diagnosed and resolved Gazebo physics and plugin issues, reducing simulation environment setup time by 40%
- Executed a full SLAM and AMCL workflow, validating autonomous navigation behavior across 20+ test scenarios

Project ARES (Autonomous Resilient Engagement System) – GNC Autonomy Engineer | *Houston, Texas* 2026 - Present

- Developed a DRL agent for 6-DOF autonomous docking, achieving a 96% mission success rate
- Implemented a Lyapunov based safety layer to ensure zero collision boundary violations during proximity operations
- Accelerated neural controllers for edge deployment on radiation hardened hardware achieving <15ms latency
- Integrated stochastic modeling to evaluate controller robustness, running 500+ Monte Carlo operation trials

Vanguard Node – *Counter UAS* | Computer Vision Engineer | *Houston, Texas* 2025 - 2026

- Architected a deterministic C++/Python ML systems stack orchestrating 4 real-time streams with 99.9% timing reliability
- Developed embedded-optimized inference pipelines using 8-bit quantization, reducing model footprint 4×
- Engineered a sensor fusion pipeline utilizing Kalman filtering to target track, moving at 2m/s with a 95% confidence
- Designed tracking using EKF achieving <0.8° angular error, 3 cm positional drift, and 94% track continuity

PROFESSIONAL EXPERIENCE

Julia – *Research Lab* | Computer Vision Engineer | *Houston, Texas* 2024 - Present

- Awarded NSF grant funding to build and validate a computer-vision-powered platform used in live operator pilots
- Integrated multi-camera inputs and calibrated camera rigs to support consistent cross-view defect detection
- Designed and deployed an end-to-end visual perception pipeline processing 5,000+ images per batch

Lockheed Martin – *Space Defense Tech* | AI Research Engineer Intern | *Houston, Texas* 2025 - 2025

- Developed an LLM-powered document assistant using embeddings and retrieval pipelines (RAG) to enable queries
- Engineered AI agents and supporting pipelines in a Linux environment using C++/Python

Work Eligibility | U.S. Citizen | Eligible for U.S. Top Secret Clearance